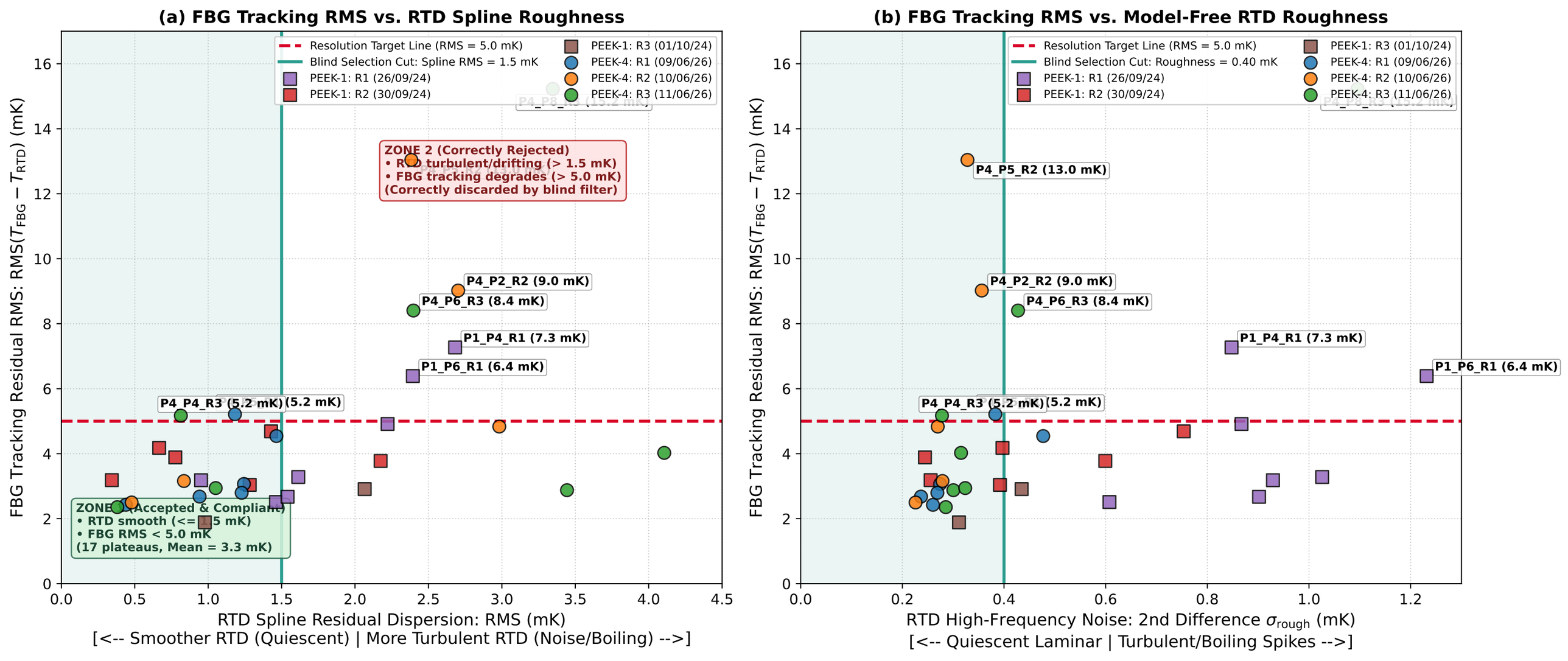
(a) FBG Tracking RMS vs. RTD Spline Roughness
Resolution Target Line (RMS = 5.0 mK)
Blind Selection Cut: Spline RMS = 1.5 mK
PEEK-1: R1 (26/09/24)
PEEK-1: R2 (30/09/24)
PEEK-1: R3 (01/10/24)
PEEK-4: R1 (09/06/26)
PEEK-4: R2 (10/06/26)
PEEK-4: R3 (11/06/26)
ZONE 2 (Correctly Rejected)
• R2 turbulent/drifting (> 1.5 mK)
• FBG tracking degrades (> 5.0 mK)
(Correctly discarded by blind filter)
P4_P2_R2 (9.0 mK)
P4_P6_R3 (8.4 mK)
P1_P4_R1 (7.3 mK)
P1_P6_R1 (6.4 mK)
P4_P4_R3 (5.2 mK)
(5.2 mK)
ZONE 1 (Accepted & Compliant)
• RTD smooth (<= 1.5 mK)
• FBG RMS < 5.0 mK
(17 plateaus, Mean = 3.3 mK)
FBG Tracking Residual RMS: RMS($T_{FBG} - T_{RTD}$) (mK)
RTD Spline Residual Dispersion: RMS (mK)
[<-- Smoother RTD (Quiescent) | More Turbulent RTD (Noise/Boiling) -->]
(b) FBG Tracking RMS vs. Model-Free RTD Roughness
Resolution Target Line (RMS = 5.0 mK)
Blind Selection Cut: Roughness = 0.40 mK
PEEK-1: R1 (26/09/24)
PEEK-1: R2 (30/09/24)
PEEK-1: R3 (01/10/24)
PEEK-4: R1 (09/06/26)
PEEK-4: R2 (10/06/26)
PEEK-4: R3 (11/06/26)
P4_P5_R2 (13.0 mK)
P4_P2_R2 (9.0 mK)
P4_P6_R3 (8.4 mK)
P1_P4_R1 (7.3 mK)
P1_P6_R1 (6.4 mK)
P4_P4_R3 (5.2 mK)
(5.2 mK)
FBG Tracking Residual RMS: RMS($T_{FBG} - T_{RTD}$) (mK)
RTD High-Frequency Noise: 2nd Difference σ_{rough} (mK)
[<-- Quiescent Laminar | Turbulent/Boiling Spikes -->]